

PSYCHOLOGY PRINCIPLES USED IN UI DESIGN

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INTRODUCTION

User experience design is an interdisciplinary subject, that draws knowledge from a variety of fields. One of them, probably the most fundamental, is psychology. Understanding how people think and behave is, after all, the key to providing them with good user experience that fits their needs and leaves them satisfied with products or services.

However, five thousand words are not enough to cover the entire plethora of psychological principles and insights used in user experience design. Therefore, here I would like to cover the main principles used in user interface design, an essential part of user experience design – without claiming to be exhaustive. Hence, I will not talk about psychological principles used during other phases of the design process, such as bias or heuristics at play when testing users. For a review of psychology principles used in user experience design see Kohler (2024) or [Growth Design website](#).

MEMORY

Miller's Law: the magical number seven, plus or minus two

Miller (1956), in his famous paper “The magical number seven, plus or minus two: Some limits on our capacity for processing information”, showed that we are able to remember seven plus or minus two “chunks” of information in our short-term memory.

Nielsen Norman Group (NNgroup, 2021) argues that the magical number is extremely helpful for user experience design, but not in the way we generally think of it. When looking for articles online that explain Miller's Law, one of the most widely suggested applications is to reduce the number of items in the website header. Although in many cases this is certainly more user-friendly, it has little to do with Miller's magical number, as you do not have to remember those header items. Instead, Nielsen Norman Group suggests that this law is relevant when creating discount codes or similar elements that users actually have to remember. *Revolut*, for instance, is one of the few banks that chunk their one-time passwords (OTP) into two groups of three digits separated by a hyphen, instead of a unique group of six.

Working memory, external memory, and recognition rather than recall

Although Miller's Law is relevant for user experience design, most of the time the users' task is not retaining information in their short- or long-term memory¹ but rather keeping

¹ However, it should be *our* goal as designers to make sure that users remember our website, application, or story, as explained by Soegaard (2025).

that information in mind to complete a task. Here is where the concept of working memory enters the scene. Baddeley and Hitch (1974) described this type of memory as composed of a visuospatial notebook and a phonological loop, controlled by a central executive. Working memory allows us to remember some information to manipulate it.

Budiu (2018) describes it as “a buffer or scratchpad in which the mind deposits information relevant to the current task”. The author explains that putting a high burden on working memory implies a high cognitive load. Since we do not want the users to perceive cognitive load, the design should support users through external memory, described as “any tool or UI feature that allows users to explicitly save and access information needed during a task” (ibidem). For instance, many websites sell similar items, which might differ from one another for some details. Comparison tables, which allow users to compare items more easily, are a perfect example of external memory. Figure 1 illustrates this concept with the Italian bank *Hype*: the pricing option “Hype Premium” has all the features of the “Hype Next” option plus some others, and the same applies to the “Hype Next” option with respect to the “Hype” option. This way, users can easily see all the options a pricing plan has.

A ognuno il suo HYPE

Nessuna preoccupazione e nessun costo nascosto.
Scegli il conto più adatto a te, al resto ci pensa HYPE.



HYPE

Il conto a canone zero per accreditare il tuo stipendio, senza limiti di ricarica e con bonifici istantanei gratuiti.

Gratis

Apri conto HYPE

Vantaggi del conto:

- Carta virtuale**
Gestisci spese e pagamenti con la carta virtuale dal tuo smartphone
- Bonifici**
Invia bonifici ordinari e istantanei gratuiti
- Prelevi**
Preleva gratis fino a 250€ al mese da ogni sportello
- Ricarica del tuo conto**
Aggiungi soldi sul tuo conto senza limiti
- Cashback**
Ottieni fino al 10% di cashback dai tuoi acquisti online⁴
- Comparatore Facile.it in app**
Risparmia su bollette, assicurazioni, prestiti e noleggio a lungo termine con Facile.it⁵

Hai meno di 18 anni?
Puoi aprire il conto HYPE nella sua versione <18, al compimento dei 18 anni il tuo conto diventerà automaticamente un conto HYPE a tutti gli effetti.

Apri conto HYPE

IL PIÙ AMATO



HYPE Next

Il conto HYPE con in più assicurazioni incluse, prelievi senza commissioni e una carta fisica gratuita.

2,90 € / mese

Apri conto HYPE Next

Vantaggi del conto:

- Con Next hai tutti i vantaggi del conto HYPE**
Carta, app, prelievi gratis, bonifici gratis, ecc.
- Carta di debito Mastercard**
Acquista in Italia e all'estero, online e in negozio con una carta fisica
- Prelievi senza commissione**
Preleva in Italia senza costi da qualsiasi sportello
- PagoPA / CBILL / Bollo auto**
Gratuiti per tutto il 2025²
- Collega conti**
Aggrega tutti i tuoi conti in HYPE per un controllo totale dei tuoi soldi
- Polizza acquisti e prelievi**
Rimborso su differenza prezzo, acquisti danneggiati o furti negli ATM³
- Assicurazione elettrodomestici**
Assistenza per riparazione elettrodomestici, TV, console e altri sistemi di intrattenimento³
- Assicurazione medica**
Assistenza per consulto medico telefonico³



HYPE Premium

Il conto HYPE Next con in più assistenza prioritaria, zero commissioni su PagoPA e tanti vantaggi per chi viaggia.

9,90 € / mese

Apri conto HYPE Premium

Vantaggi del conto:

- Tutti i vantaggi dei conti HYPE e HYPE Next**
HYPE Web, ricarica con carta gratis e molto altro
- Paghi e prelievi gratis nel mondo**
Ovunque tu sia, prelevi e paghi in qualsiasi valuta senza commissioni
- Carta World Elite Mastercard**
Richiedila gratis e ottieni servizi esclusivi come assistenza, Lounge e Fast Track aeroportuali
- Assistenza prioritaria**
Ricevi supporto via WhatsApp, email, telefono, chat
- Assicurazioni Premium**
Tutte le assicurazioni incluse nel conto HYPE Next più le coperture su viaggi, spese mediche all'estero, tutela legale sul cyber risk e assistenza digitale³
- Paga bollette e bollettini**
Puoi pagare in app Premarcati, in bianco, MAV e RAV, PagoPA e cbill senza costi aggiuntivi

Figure 1 - *Hype* pricing plans allow easy comparisons

Our limited capacity (and willingness) to remember is the core motivation of one of Nielsen's usability heuristics (Nielsen, 2024), which is *recognition rather than recall*. Since users visit a lot of websites and use a lot of applications, expecting them to remember every aspect of our design is simply unrealistic. For this reason, the design should make users recognize familiar elements (icons, types of interactions, and so on) rather than using innovative and very distinctive ones that users must remember in order to interact with the website. Hence, it is always better to add verbal descriptions to icons. Budiu (2024) explains that a classical example of this heuristic is a menu system: almost every action on a menu system could be done through a keyboard shortcut, but there are rarely used functions that we might want to use through a menu system, where every action is visible. This is also a perfect example of another of Nielsen's usability heuristics: *flexibility and efficiency of use*, which means providing different methods to complete an action so that experts and novices can both use the system according to their level of expertise. Figure 2 represents the menu system of *Microsoft Word*.



Figure 2 - *Microsoft Word* and the recognition rather than recall heuristic in the ribbon bar

INFORMATION PROCESSING AND DECISION MAKING

System 1 and System 2

Since applications and websites are full of information to process, it is essential to note that people can elaborate information and make decisions based on two systems: System 1 and System 2 (Kahneman, 2011; the terms were actually coined by Stanovich and West, 2000), also called “automatic system” and “reflective system” by Thaler and Sunstein (2021). System 1, or automatic system, is what Kahneman refers to when he talks about “thinking fast”: fast, automatic (without control or will), often unconscious, emotional and effortless. For instance, recognizing a face, or understanding that someone is annoyed because of their voice. On the contrary, System 2, or reflective system, is what Kahneman describes as “thinking slow”: slow, deliberate, conscious, analytical and it requires a lot of effort. For instance, understanding logical arguments, or solving mathematical problems such as multiplying seventeen by twenty-four.

As pointed out by Acevedo (2025), in most cases, we should want our websites and applications to accommodate users' System 1 thinking: they should meet users' mental models, and they should speak users' language. Nonetheless, this is not always the case. Sometimes System 2 is not only unavoidable, but also desirable. When completing difficult tasks, users must use System 2. Hence, designers' work should be aimed at accommodating users'

System 1 when possible, and at supporting them when they have to focus on something, like comparing different products, that requires System 2.

Fiscozen is an Italian company that helps freelancers manage their taxes, an activity typically handled by System 2, through captivating aesthetics and an easy dashboard. Thus, the company can also persuade casual visitors that are using their System 1.

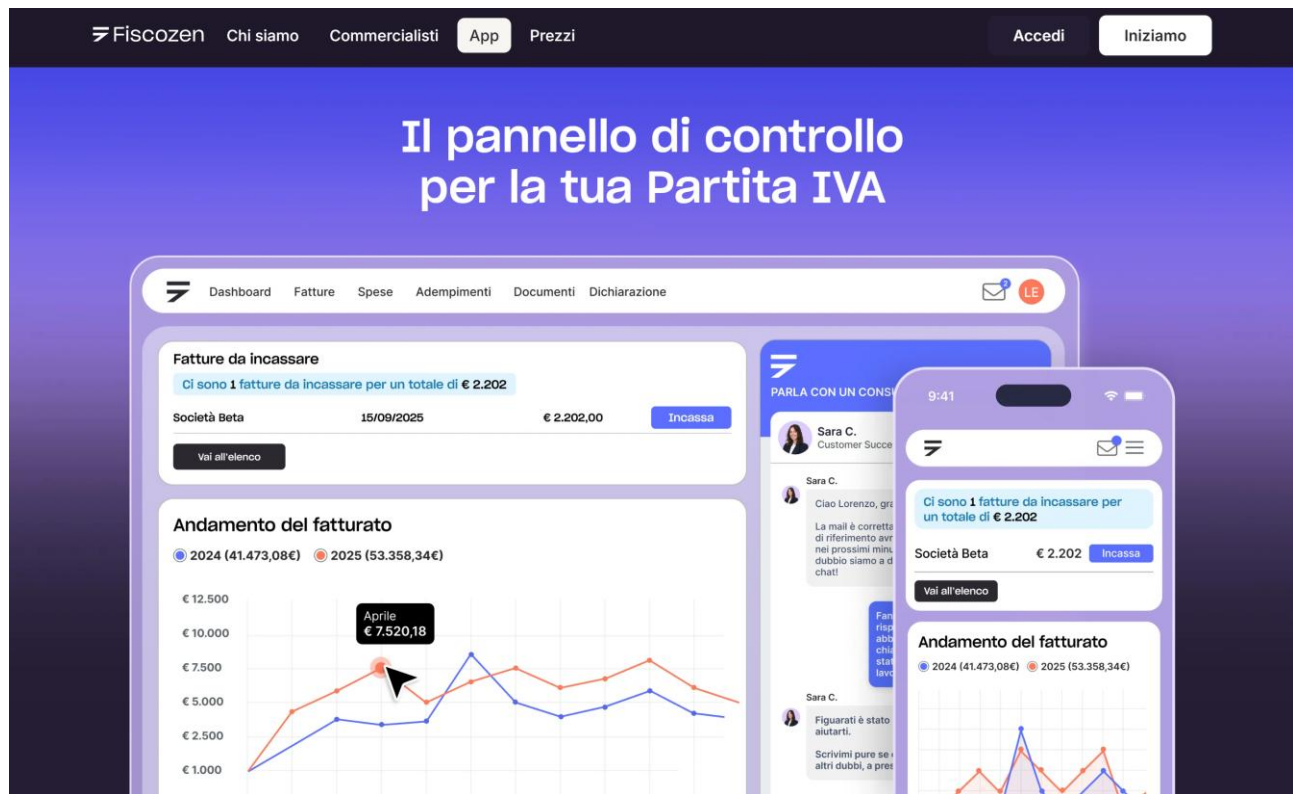


Figure 3 - *Fiscozen* UI accommodates both System 1 and System 2

Prospect theory

Doubtlessly, prospect theory (Kahneman and Tversky, 1979) is one of the most fundamental and influential theories about decision making. It was formulated as an alternative – or at least complementary – model to the expected utility model, widely known in economics, in the context of decision making under risk.

The theory postulates that people perceive monetary outcomes as gains and losses with respect to a neutral reference point, which “can be affected by the formulation of the offered prospects, and by the expectations of the decision maker” (ibidem, p. 274). Furthermore, they are willing to avoid risk in choices involving sure gains and to seek risk in choices involving sure losses, resulting in the concept of “loss aversion”.

For instance, when people have to choose between a sure gain of nine hundred dollars or a ninety percent probability of gaining a thousand dollars, they opt for the sure gain, even if they could win more, avoiding risk. However, when they have to choose between a sure loss

of nine hundred dollars or a ninety percent probability of losing a thousand dollars, they opt for the probability of losing more, seeking risk (Kahneman, 2011). As stated by Harley (2016), “the pain of losing is greater than the satisfaction of an equivalent gain”.

Harley (ibidem) recommends giving sure rewards such as coupons instead of possible rewards, like sweepstake tickets. Another suggested application is framing the same facts in a positive way, instead of a negative way: a ninety-five percent satisfaction rate is better than a five percent complaint rate, even if they refer to the same evaluation. *NordVPN* highlights that choosing a two-year long subscription to its service could make the user save seventy-three percent instead of highlighting that the user would pay twenty-three percent of the original price. The company frames the same fact as avoiding loss, rather than getting a gain.

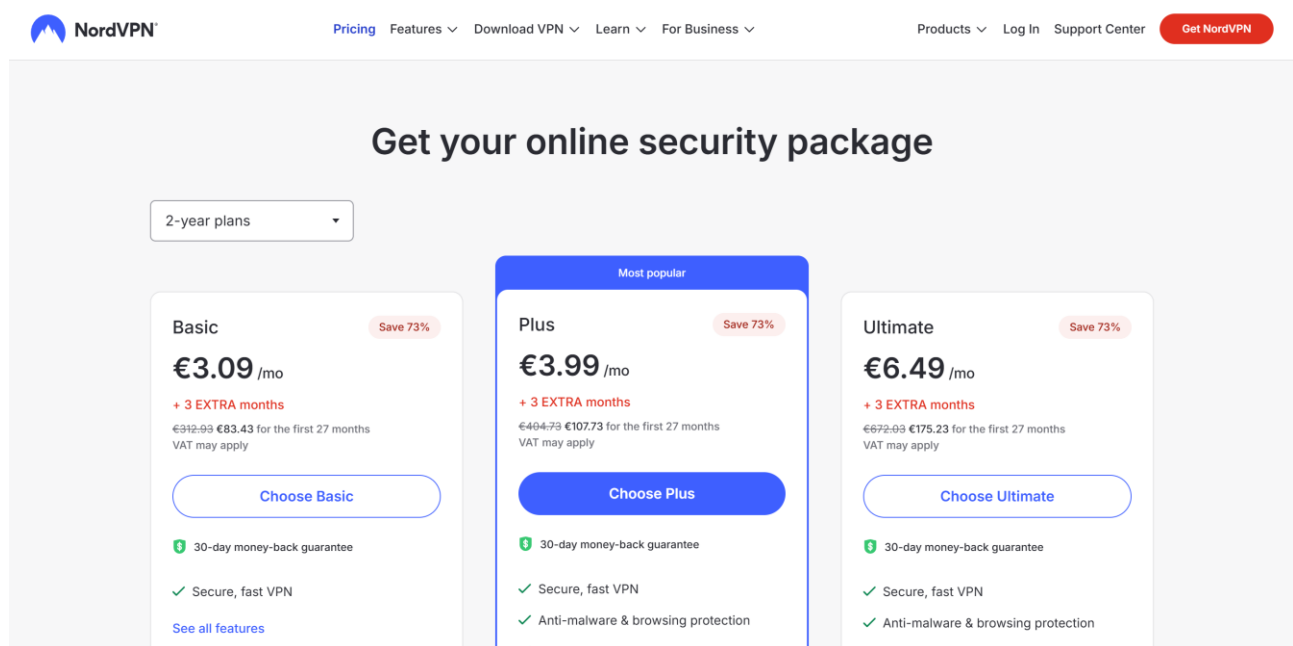


Figure 4 - *NordVPN* and loss aversion

Another consequence of prospect theory is the “endowment effect” (Kahneman, 2011), which is our tendency to evaluate as more valuable the things we own, even if we have just acquired them. This effect is at work whenever we are using a free trial of a product or a service. Given that we have been using that product or service for a month, we might start to think that we own that, therefore perceiving the end of the free trial as a loss. Loss aversion and endowment effect could convince us to buy that product or service.

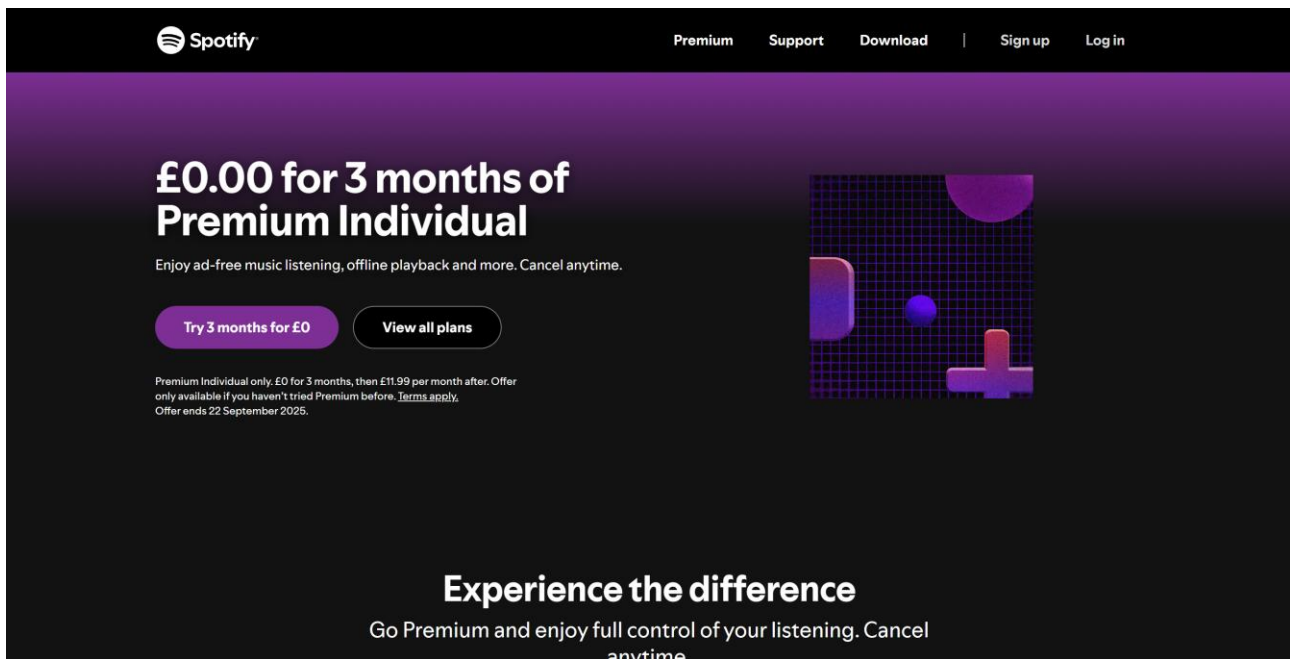


Figure 5 - [Spotify](#) free trial

Choice architecture, nudging, and decoy effect

Choice architecture and nudging

Decision making is also influenced by some contextual elements, such as items order. Thaler and Sunstein (2021) called it “choice architecture” and argued that it influences people’s decision-making process. For instance, in their famous book “Nudge: the final edition” (ibidem) they give an example of how the order of the dishes in a scholastic canteen could lead to an increase in vegetable consumption.

It is important to note that when designing the choice architecture, whether we are creating a restaurant menu or the search results in an e-commerce, we are not compromising user’s freedom of choice but rather influencing it. Whenever this influence benefits the users’ current or future health or wellness, the authors call it “nudge”, as it represents a gentle and paternalistic push to what’s better for them.

A classic application of choice architecture is the design of default options. For instance, a company could set the most environmentally friendly option as default option or put it first in a list of options. The Interaction Design Foundation (2017) showed how Google Flights nudges users by ordering flights based on their carbon footprint, showing the least environmentally impactful first.

An example of default option is *Netflix* autoplay button, which automatically skips to the next episode of a tv series. I’d argue that leading you to do binge-watching is not, usually, pursuing your wellbeing. Therefore, I would classify this as an example of choice architecture, and not of nudge.

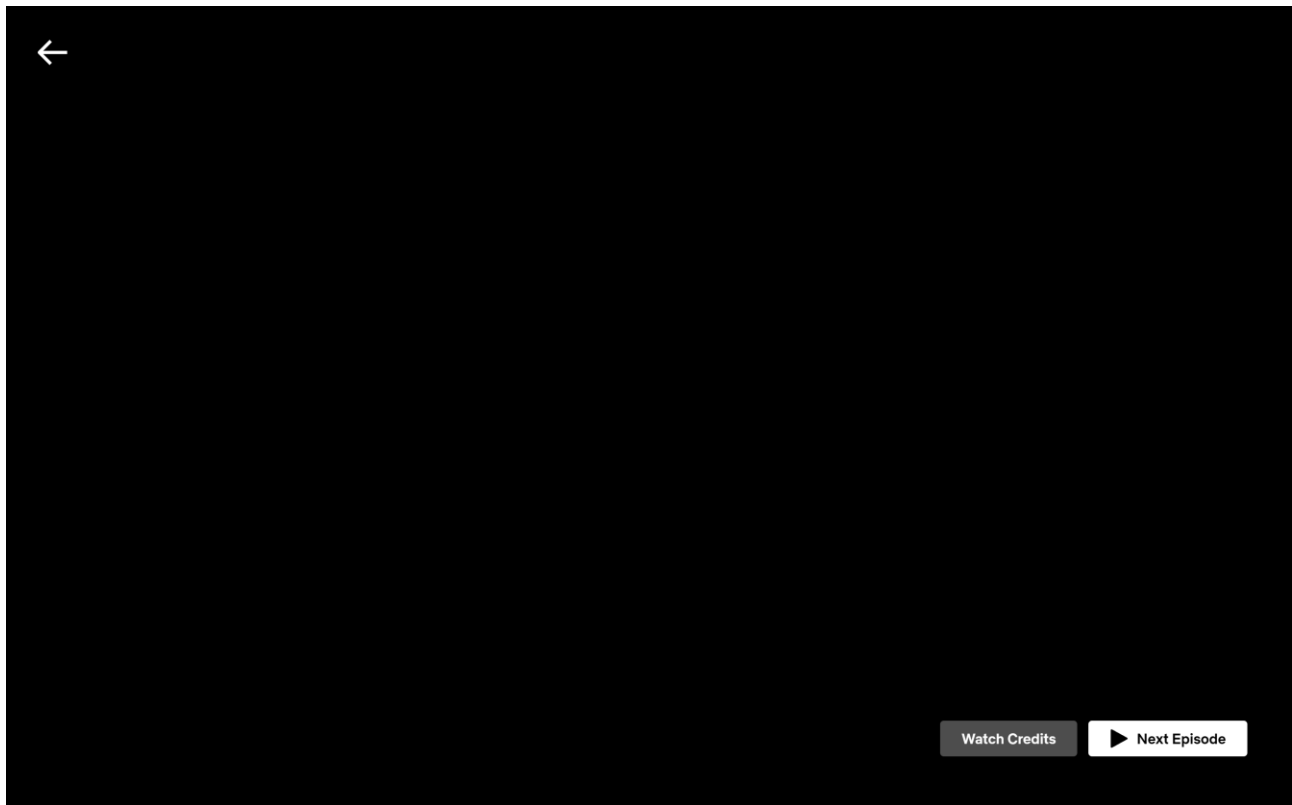


Figure 6 - [Netflix](#) auto-play button as a default option

Decoy effect

Another element that impacts decision in economic contexts, especially pricing plans, is the asymmetric dominance effect, introduced by Huber, Payne, and Puto (1982). Also called “decoy effect”, this phenomenon explains how adding a third asymmetrically dominated option (called “decoy”) could change the preference between two options. An option is a decoy if “it is dominated by at least one alternative in the set but is not dominated by at least one other” (ibidem, p. 90).

An excellent example of this is the pricing strategy adopted by the magazine *The Economist*, as explained by Dan Ariely in a TED Talk conference (Ariely, 2008). The magazine offered a year-long subscription to their website for fifty-nine dollars, a year-long subscription to their website and print edition for one hundred and twenty-five dollars (the target option), and finally a year-long subscription to the print edition only, for one hundred and twenty-five dollars (the decoy option). Clearly, the third option (the decoy) is not convenient if compared to the second (the target). However, its presence influence people’s perception of the target option, resulting in more people choosing *The Economist*’s preferred subscription plan. Ariely (ibidem) explains that, in an experiment with some MIT (Massachusetts Institute of Technology) students, removing the decoy would bring the target option from being the most preferred option to being the least preferred option.

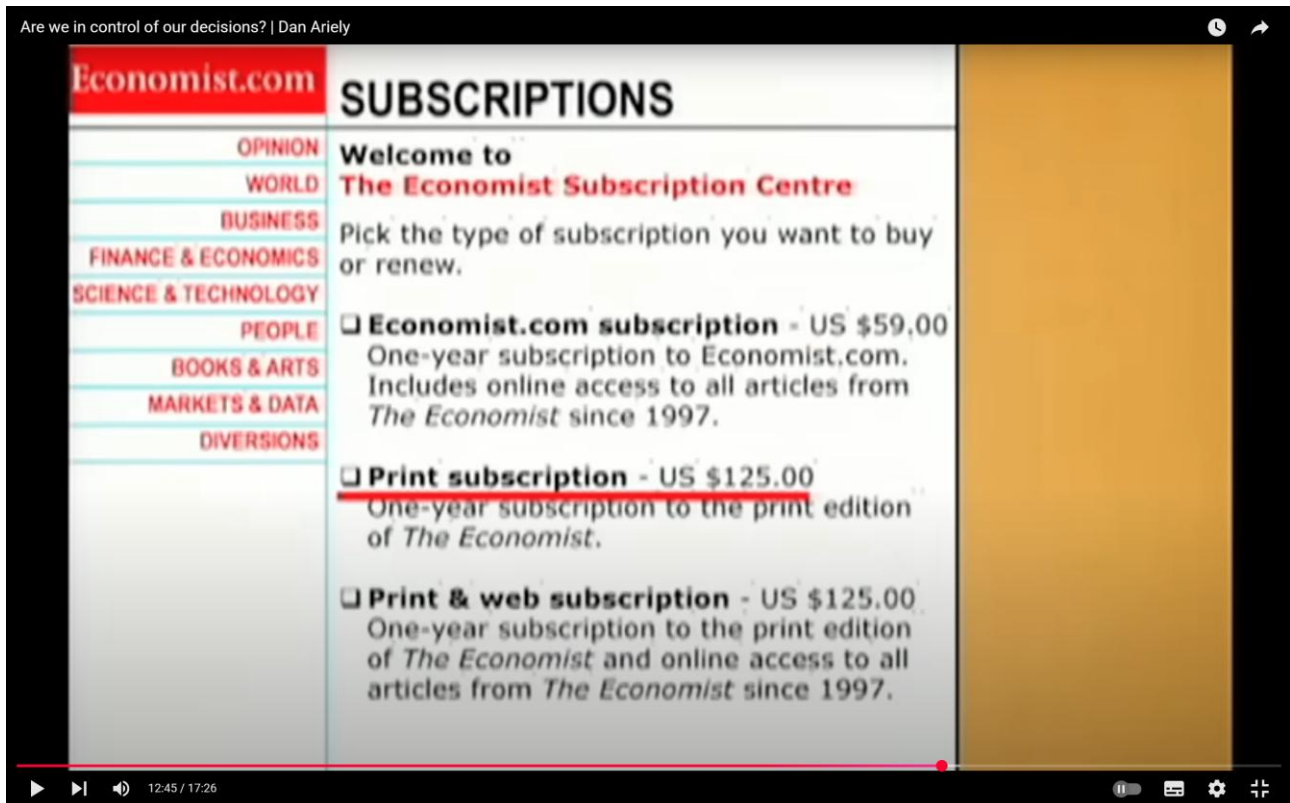


Figure 7 - *The Economist* and the decoy effect (via [Dan Ariely's TED Conference](#))

Choice overload and Hick's Law

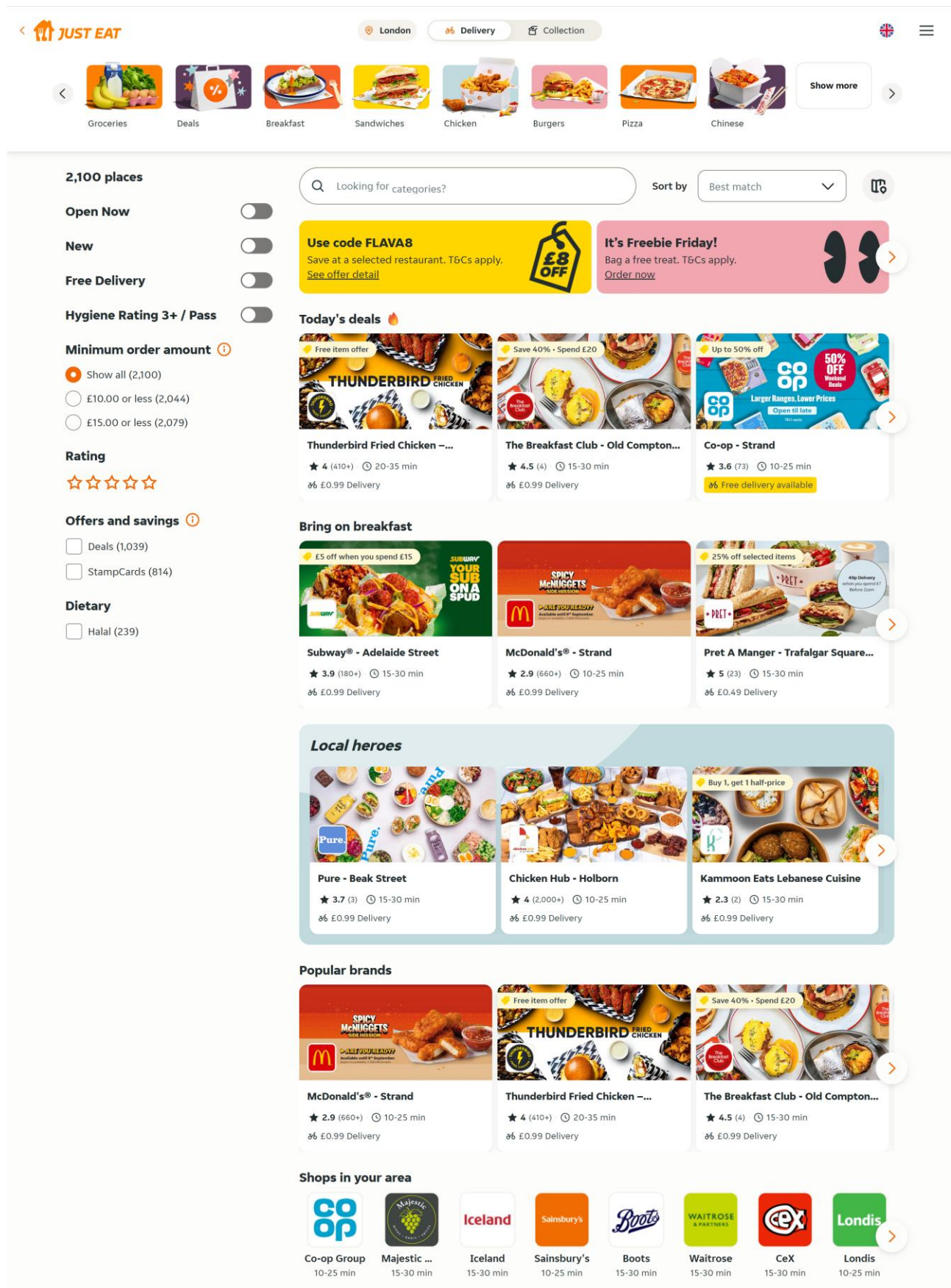
Choice overload

When reasoning about choice, it is worth noting that having a lot of options is not always optimal, a concept often referred to as “choice paradox”. Chernev, Böckenholt, and Goodman (2015), who wrote a meta-analysis to review the scientific literature concerning this topic, talk about the term “choice overload” or “overchoice”, which “is typically used in reference to a scenario in which the complexity of the decision problem faced by an individual exceeds the individual’s cognitive resources” (ibidem, p. 335).

According to the model proposed by these authors, there are four key factors of choice overload: choice set complexity, decision task difficulty, preference uncertainty, and decision goal. Therefore, choice overload is likely to happen whenever choosing is complex (because there is not any dominant option or options are not perceived as of high quality), there are some context-dependent factors (such as time constraints, decision accountability, number of attributes describing each option, and presentation format), people do not have certain preferences (because they are not experts), and their goal is buying something (and not simply browsing).

The good news is that reducing choice overload is simple: it is enough to reduce options that are not very different from one another. However, this is not always possible. Nielsen

Norman Group (NNgroup, 2020) suggests, whenever it is not possible, categorizing options in meaningful ways, so that users could more easily browse before the offerings, or highlighting the top products or services is another way. Lastly, comparison tables lessen the burden that users must face to compare different products. For instance, *Just Eat* packed its offerings in groups such as “Today’s deals” or “Bring on breakfast”, as shown in Figure 8.

Figure 8 - *Just Eat* avoids choice overload by categorizing restaurants

Hick's Law

Choice overload is also partially overcome by applying Hick's Law in user interfaces, to make users' choices easier. Hick (1952), applying Shannon's and Weaver's information theory (as cited by Hick, 1952) to Merkel's (as cited by Hick, 1952) findings, discovered that reaction time (meaning the time to give an answer) increased logarithmically as the number of answer possibilities increased. This finding is generally referred to as "Hick's Law".

Institutions such as the Interaction Design Foundation (2016) suggest keeping this law in mind when designing lists of items, such as the links in a website header, to prevent users from leaving the website or the application. *Amazon* is a giant e-commerce so it is extremely complex by nature, but it should not dump its navigation system complexity to the user (like any system, actually). For example, Figure 9 shows that the tab "Categories" is divided into *Top Categories*, *Fiction*, *Nonfiction*, *Children's Books*, *Shorts*, *More Categories*. If most categories are well handled, *Nonfiction* features twenty-four different sub-categories, potentially bringing to choice overload.

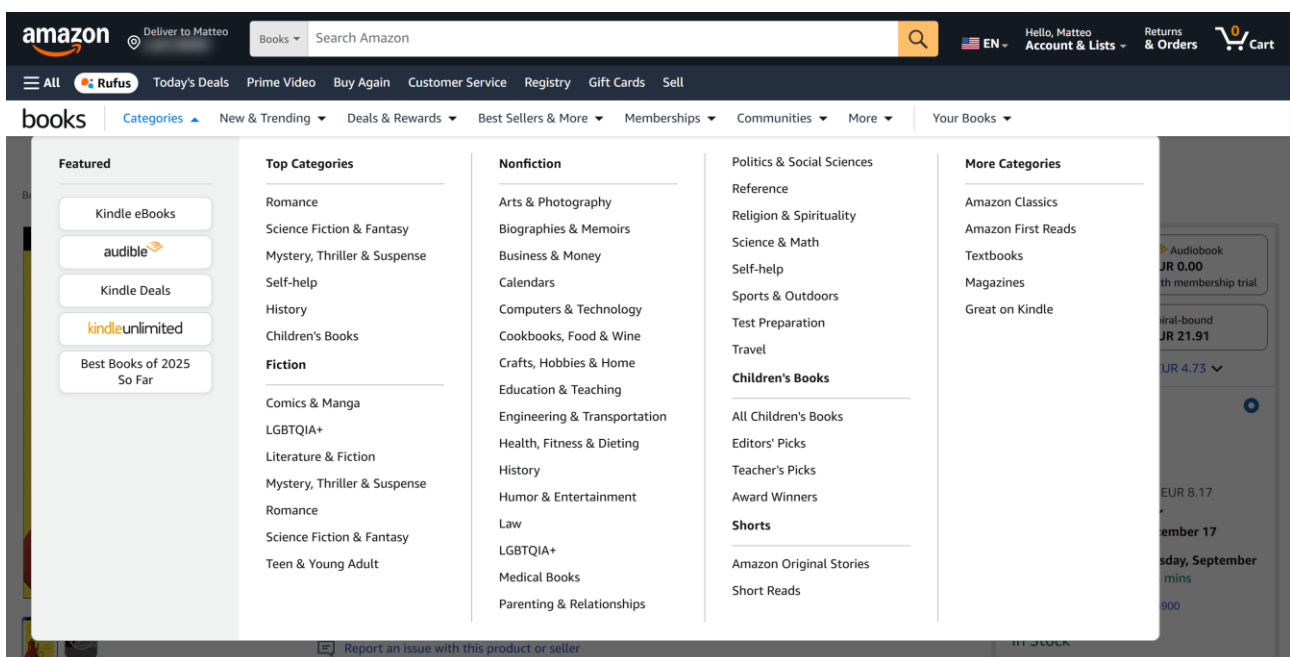


Figure 9 - [Amazon](#) and choice overload

Another useful application of Hick's Law concerns onboarding processes, as described by Soegaard (2020). Since a certain amount of complexity is unavoidable during these processes, one way of reducing its perception from the user is to simply break down the process into multiple screens with fewer options to pick from.

However, as noted by the same Interaction Design Foundation (2016), there are exceptions to Hick's Law, regarding longer menus. In fact, as reaction time is bound logarithmically to the increase of options, it matters more in shorter menus than in longer menus, but under the condition that "users already know what they want to do *before* seeing the list of choices"

(ibidem). In addition to this, Nielsen Norman Group (NNgroup, 2018) argues that long menus are usable when they meet another condition, which is that items are ordered following a specific criterion. Items are usually ordered alphabetically, but it really depends on the situation.

For instance, the Italian funeral services company *Taffo*, when opening its website, asks the user to choose the city where the service is needed. Users already know the city they are searching, so the length of the menu is not relevant.

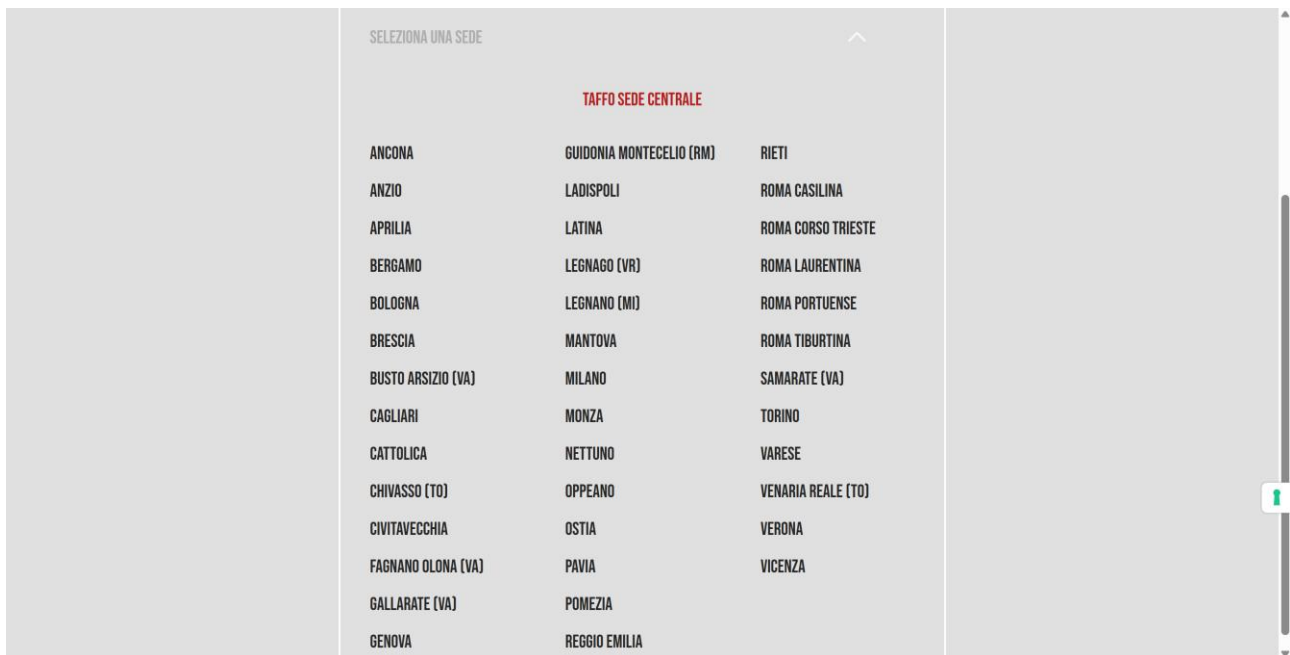


Figure 10 - *Taffo* and an alphabetically ordered long menu with known items

PERCEPTION

Gestalt laws

Some of the most fundamental perception principles come from the psychological school of Gestalt, which gave name to the homonymous laws (Interaction Design Foundation, 2016). These laws, widely used in visual design, explain how we perceive and segment elements in our visual field. It would be possible to write a whole five-thousand-word document only about Gestalt laws, but here I'll describe briefly the main ones useful for user interface design.

Closure law

The closure law states that we tend to fill in blanks between elements to perceive complete elements. This principle is often used in icon design. For instance, *Interaction Design Foundation* exploited it to communicate the percentage of people who booked their courses.

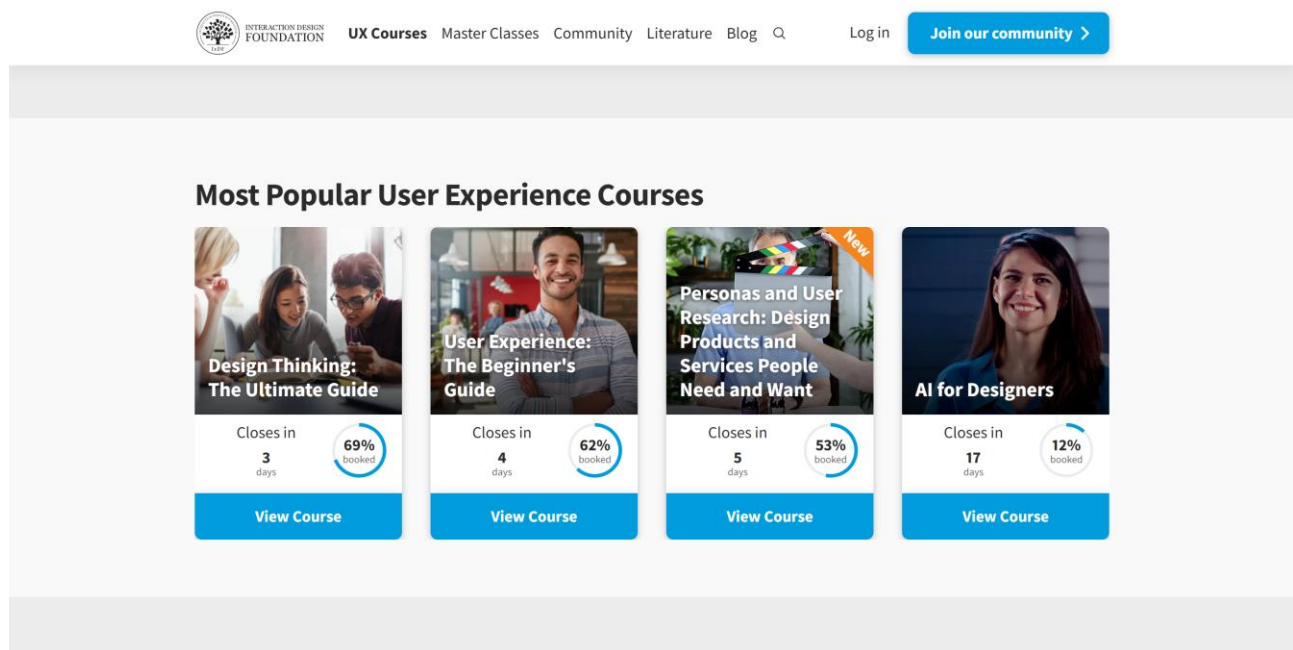


Figure 11 - [Interaction Design Foundation](#) and the closure principle

Continuity law

Our perception of obstructed elements as complete elements is explained by the continuity law. This principle is often leveraged in interface design to signal elements that do not fit the screen size, like the card in Figure 12 on *Revolut* website.

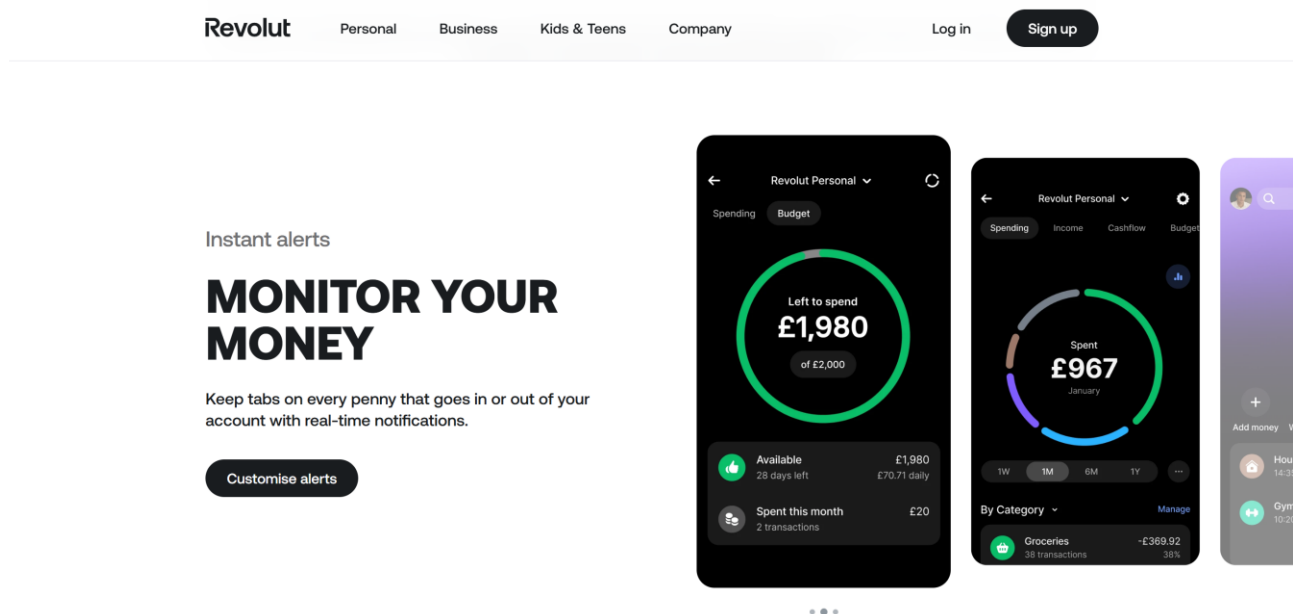


Figure 12 - [Revolut](#) and the continuity principle

Common region law

The common region law explains that we perceive elements in the same closed region as part of a whole. Designers exploit this principle to make users perceive different sections. Figure 13 shows an example on *Relume* website.

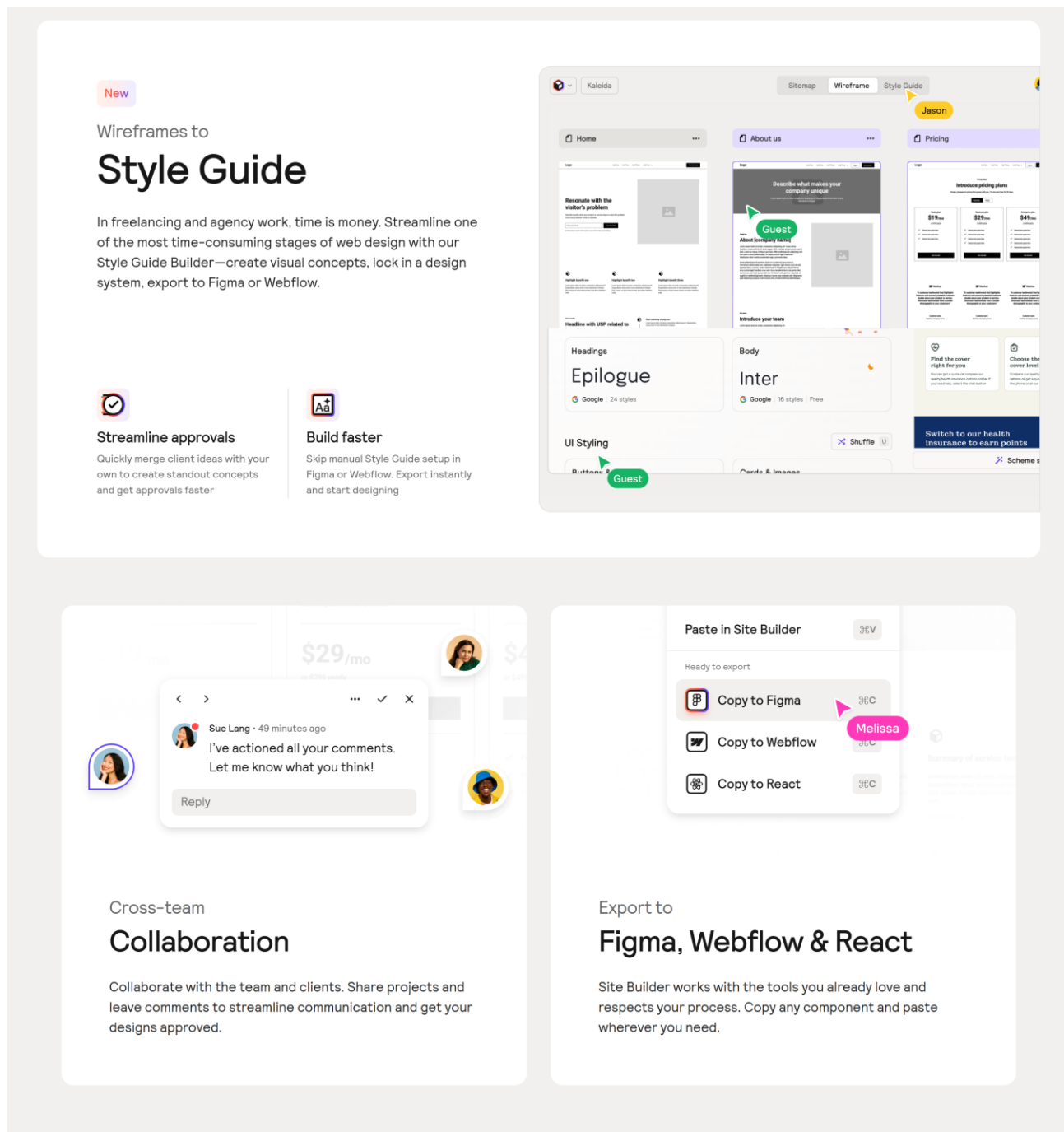


Figure 13 - *Relume* and the common region principle

Proximity law

According to the proximity law, things that are close to each other are grouped together and seen as a whole. In Figure 14, *Slides Agency* used this principle to group the image, the title, and the subject of every single news.

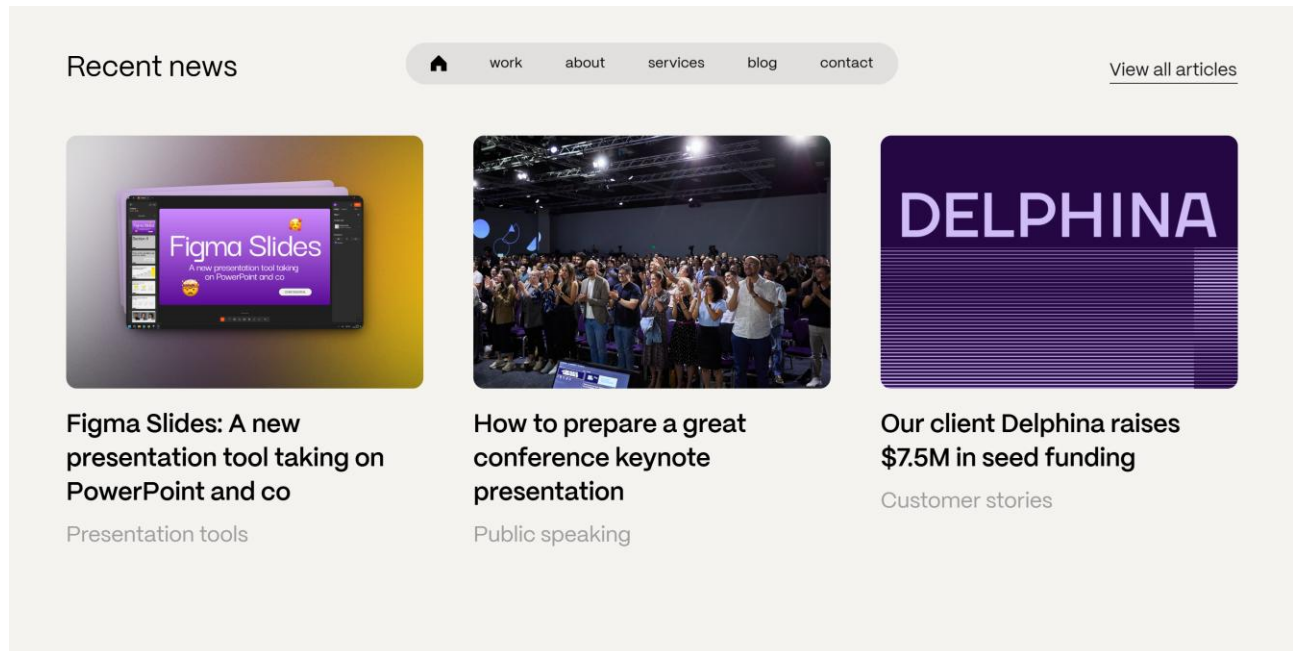


Figure 14 - *Slides Agency* and the proximity law

Figure/ground law

The figure/ground law states that we perceive things by dividing them into figures (that stand in the foreground) and their background. This law can be exploited by contrasting elements with their background, or by reducing the background opacity. For instance, Medium applies this principle when asking users to subscribe to its contributors' newsletters by lessening the background opacity, to make the pop-up stand out.

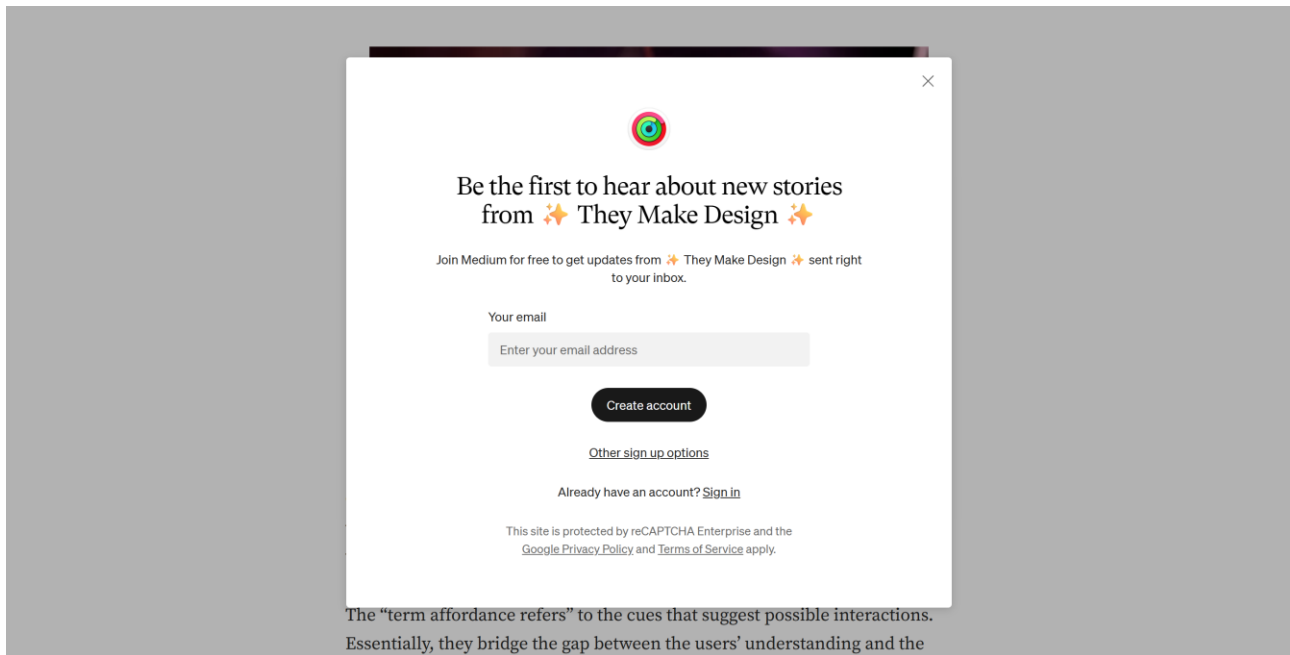


Figure 15 - [Medium](#) and the figure/ground principle

Similarity law

The similarity law explains that all other things being equal we perceive similar elements as grouped. Similarity can be achieved through elements such as color and shape. *AI Board of Innovation* used color to differentiate their equally shaped products.

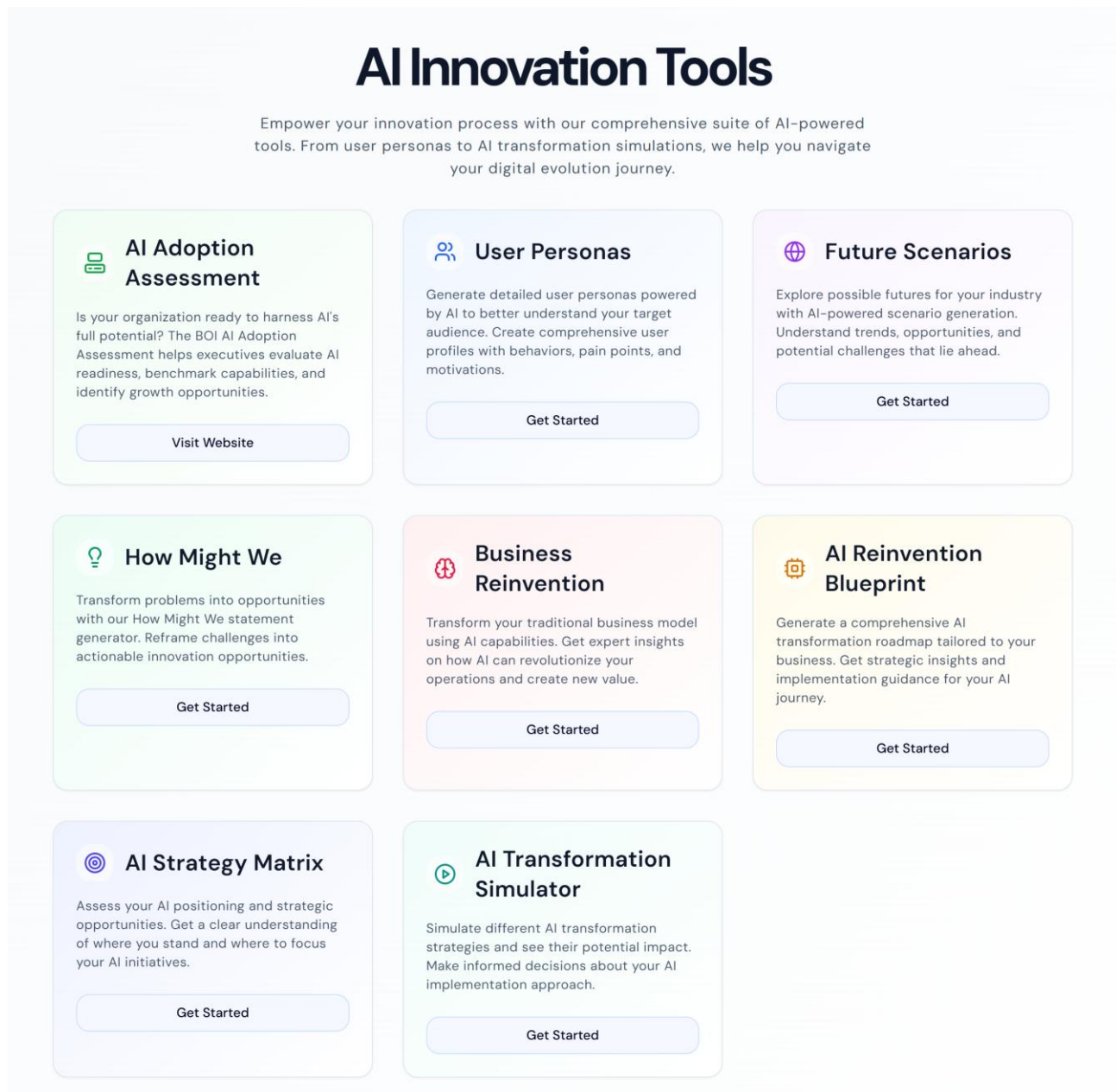


Figure 16 - *AI Board of Innovation* and the similarity principle

Common fate law

Whenever we look at a flock of birds flying in the sky or at a group of soldiers marching, we see the common fate law at work. This law states that we tend to perceive elements moving together in the same direction as a group. Web designers leverage this principle for horizontally scrolling sections, like in the example below from Kayode Sholanke's portfolio.

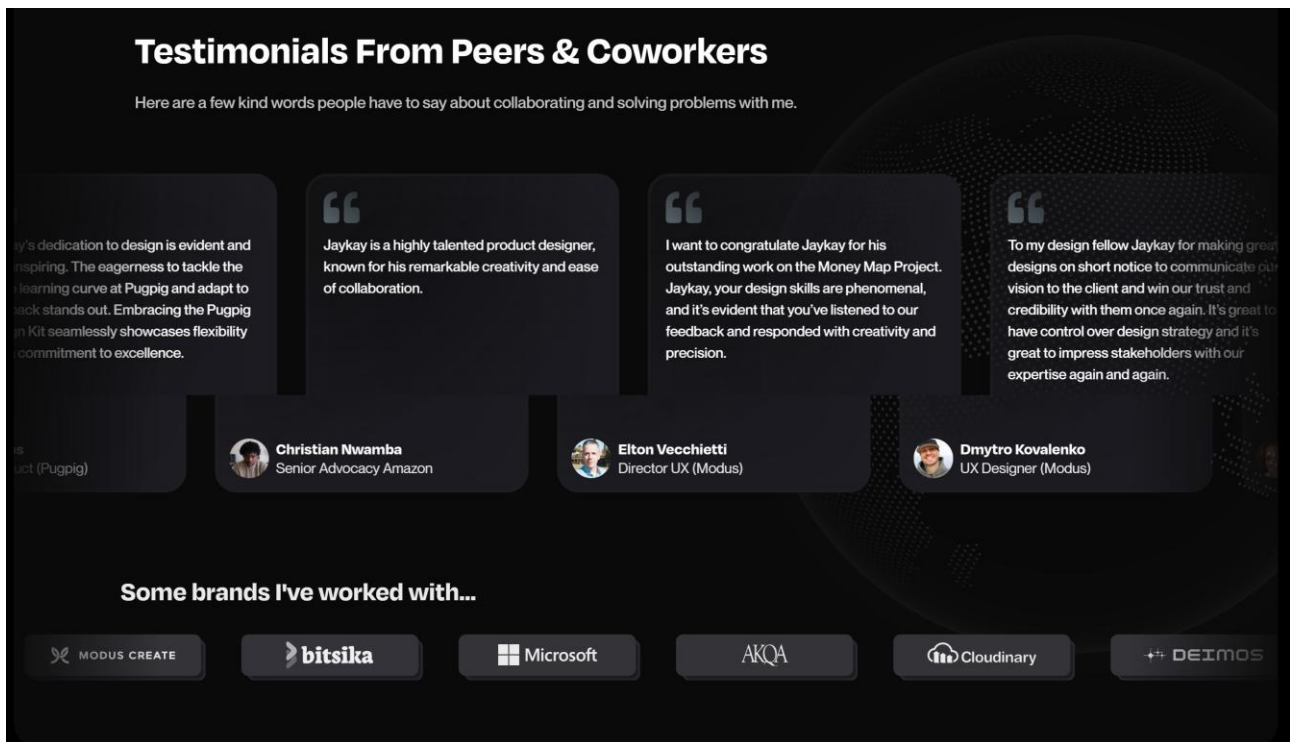


Figure 17 - [Kayode Sholanke's portfolio](#) and the common fate principle

Past experience law

The past experience law says that our perception of the world is also influenced by our past experience. This is one of the reasons why following Nielsen's (2024) "consistency and standards" heuristic is important. For instance, past experience tells us that a magnifying glass means "search", so we do not need a verbal label on a button with a magnifying glass.

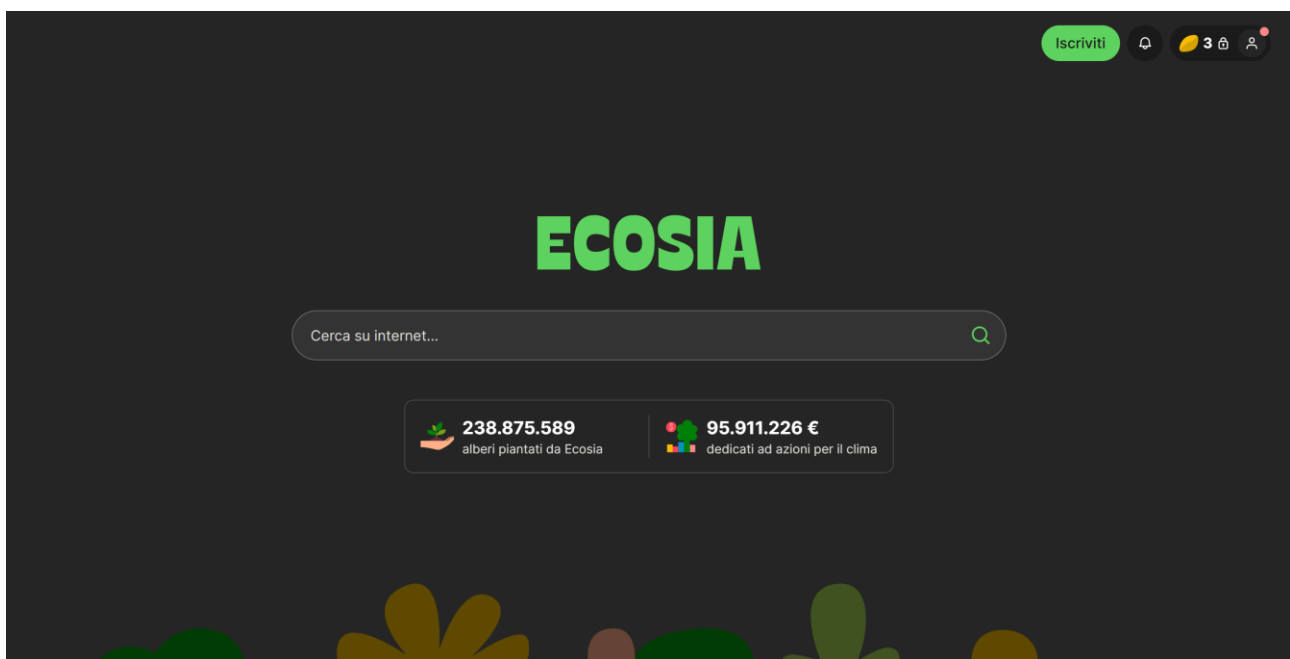


Figure 18 - Search engine [Ecosia](#) and the past experience law

Affordance and signifiers

Donald Norman borrowed the term “affordance” from James Gibson (as cited by Norman, 2019) in the first edition of his renowned book “The Psychology of Everyday Things” in 1988 (which was later renamed “The Design of Everyday Things”). Norman describes affordance as the relationship between an agent (either human, animal, or machine) and an object that tells the agent how to use the object. The author does not think of affordance as an object property, because different people might perceive the same object in different ways. A chair might be perceived by most people as something that you could sit on, or that you can move from one place to another. However, that same chair does not have those characteristics for a three-year-old child who is not tall enough to reach the chair. Besides, some affordances are perceivable, some are invisible.

Norman (ibidem) also coined the term “anti-affordance”, which is used to prevent users from interacting with an object in a wrong way (or in any way). For example, vertical black hoses placed in front of a public park are usually made of hard material. Placing rubber hoses makes it possible to both prevent cars from getting into the park (because people assume they are made of hard material) and allow authorized gardeners to enter effortlessly. This is one of the many ways through which affordances and anti-affordances can be used to prevent wrong users from interacting with a product.

Signifiers are the part of affordance that signals its presence. If affordances show *which* actions are possible, signifiers communicate *where* that actions should take place. Norman argues that, for this reason, they are more important than affordances themselves. Some signifiers are nothing but perceived affordances, like the handle of an emergency door. Signifiers can be deliberate and intentional, such as the “push” sign on a door, or unintentional. A bookmark is used to keep track of the last page we have read in a book, and it plays the role of an intentional signifier for that action. However, when a bookmark is used to check how many pages we are left with, then it plays the role of an unintentional signifier.

Interaction Design Foundation (2016) explains that every element on a touch screen has the affordance of being touched or pressed for people who were exposed to this convention before – as Norman (2013) explains, every affordance online is a perceived affordance, a consequence of a convention. Hence, signifiers are essential to indicate to users where they must press to complete a certain task. Some examples of signifiers in web and mobile design are drop shadows around a button, arrows that indicate that something is explained in another page and scroll bars that indicate that the content continues over the screen limit.

The same article suggests looking at the Interaction Design Foundation homepage. In Figure 19, the shadows around the boxes and the buttons indicate that those elements can be clicked. Furthermore, the “play” icon close to “Watch our 6-stage course creation process” reinforces the intuition that clicking that sentence will redirect you to a video content.

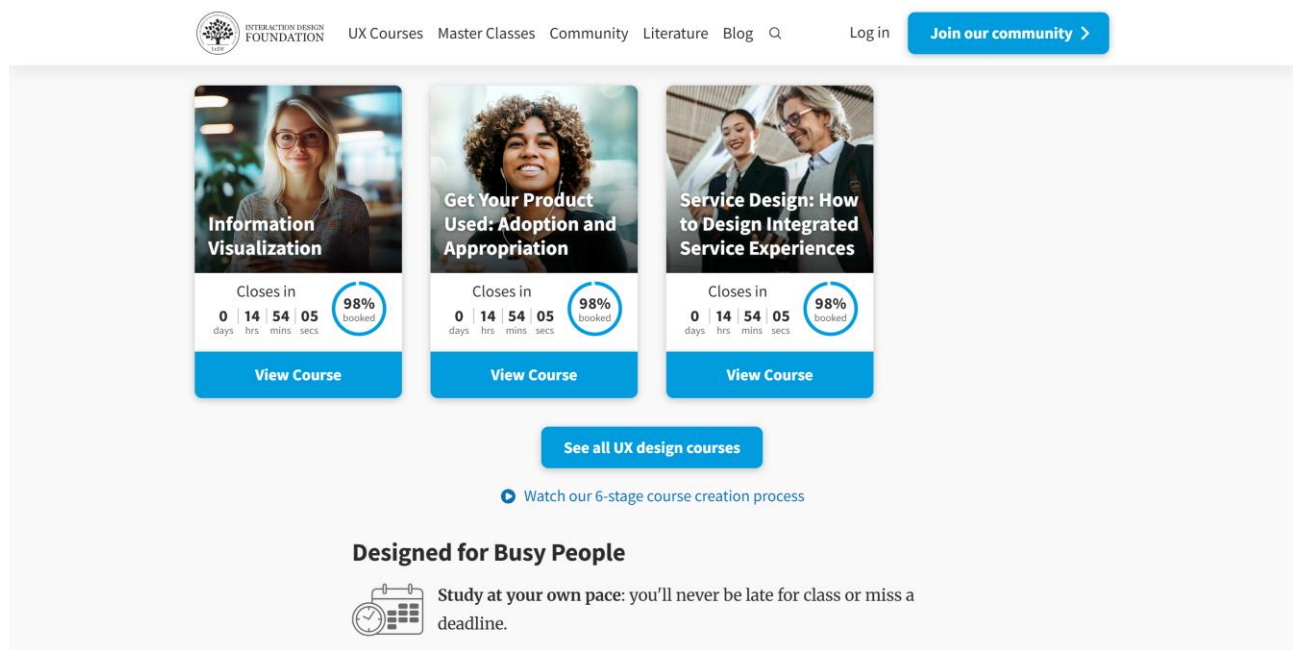
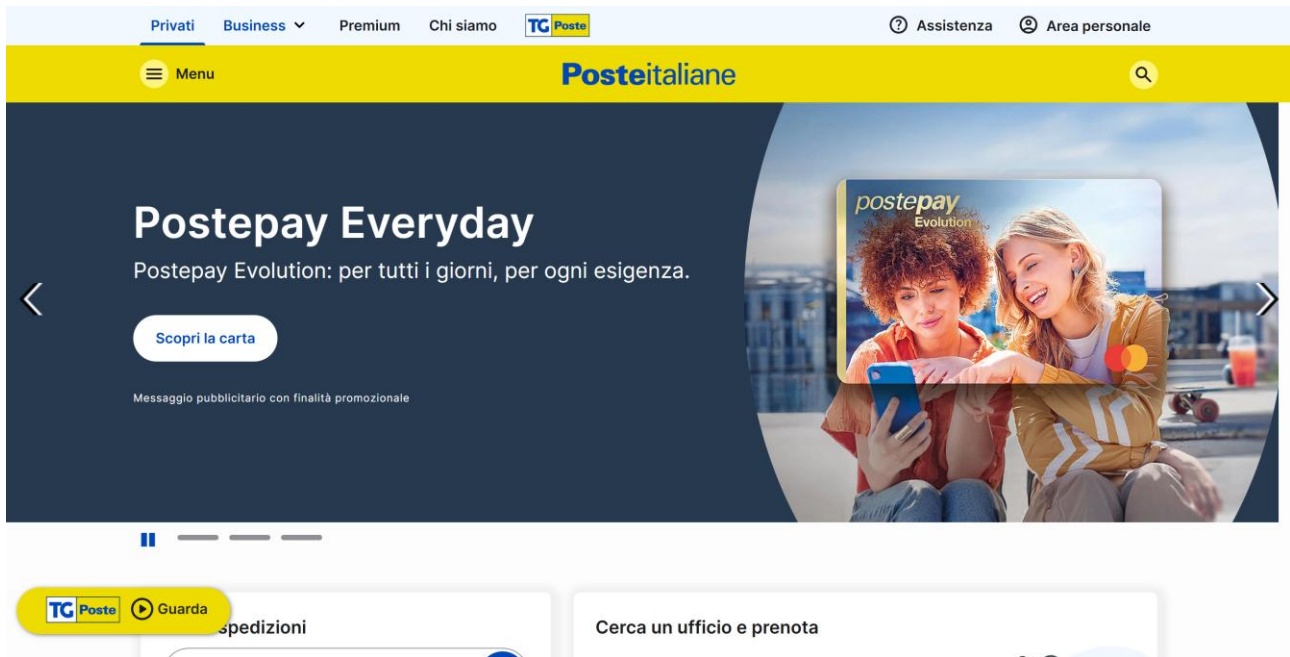


Figure 19 - [Interaction Design Foundation](#) homepage and the use of signifiers

Fitts' Law

Another relevant concept from psychology literature about perception is Fitts' Law. Fitts (1952) found that the time due to completing a motor task is logarithmically bound directly to the target's distance and inversely to its width. As explained by the Interaction Design Foundation (2016), the immediate consequence for user interface design is that buttons, or any other interactive element, must be large and close to the user's task or attention area.

As suggested by Budiu (2022), Fitts' Law could be used in menu design or wherever an element appears as the consequence of pressing or hovering over another element. Although this law might seem common sense, there are many examples of websites that do not use it, such as the Italian mail service *Poste Italiane* website below: when clicking on the "Business" tab on the header, a section appears on the left side of the page, instead of below the tab.



Business

- Professionisti e piccole imprese
- Medie e grandi imprese
- Pubblica amministrazione



Figure 20 - *Poste Italiane* does not apply Fitts' Law in his website header

SOCIAL INFLUENCE

Cialdini's principles

Since the web is an extension of our lives, it can be considered a social environment. Indeed, many insights coming from social psychology are used in web design to influence people's online behavior.

Back to [summary](#)

Social psychologist Robert Cialdini (2001), who is famous for his research on influence and persuasion, talks about six principles of persuasion. They are now widely known and used in user experience design, as well as marketing and advertising.

Reciprocity principle

The reciprocity principle states that we feel the urge to return the favor someone made us. This principle is exploited by marketing teams every time we are encouraged to take a free sample or a free trial in a shop. It plays a role in web design as well, for instance whenever we are asked to leave an email to receive a free pdf or e-book. For instance, *Growth Design* asks users to create an account to read its case studies.

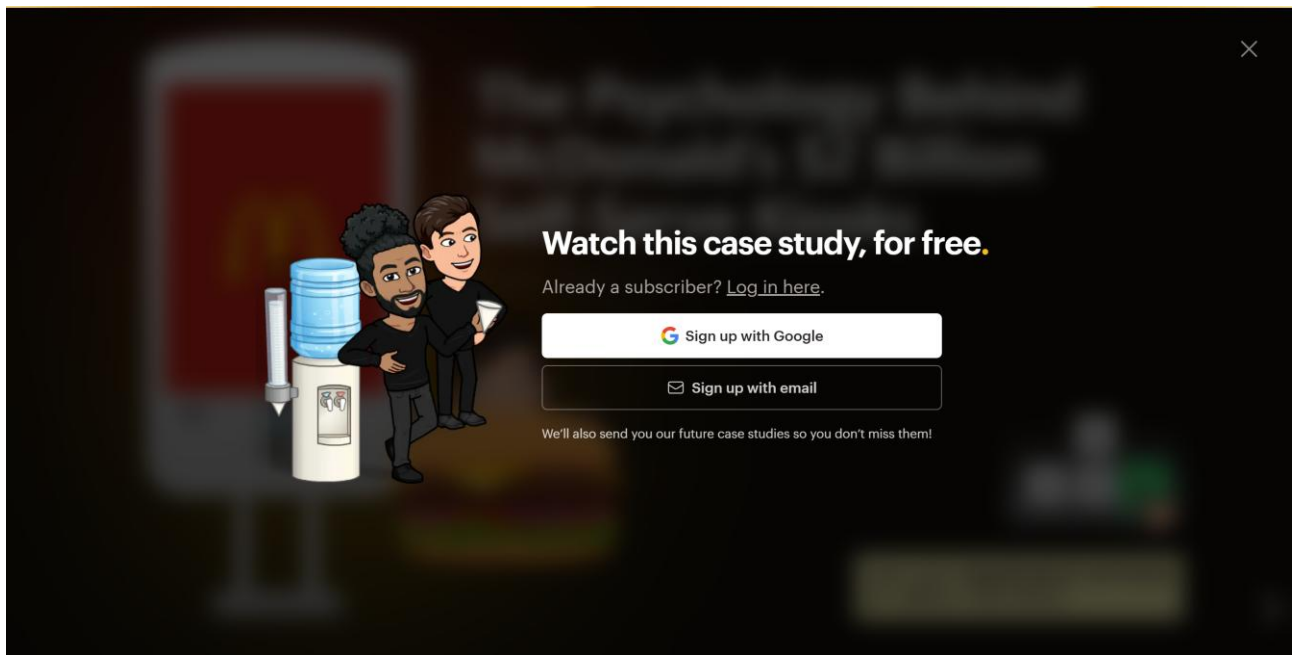


Figure 21 - *Growth Design* and the reciprocity principle

Consistency principle

The second principle, consistency, explains that people strive to appear consistent, behaving accordingly with what they say. As pointed out by [a case study from Growth Design](#), Duolingo uses this principle by asking unactive users if they still want to learn the language they have declared they wanted to learn.

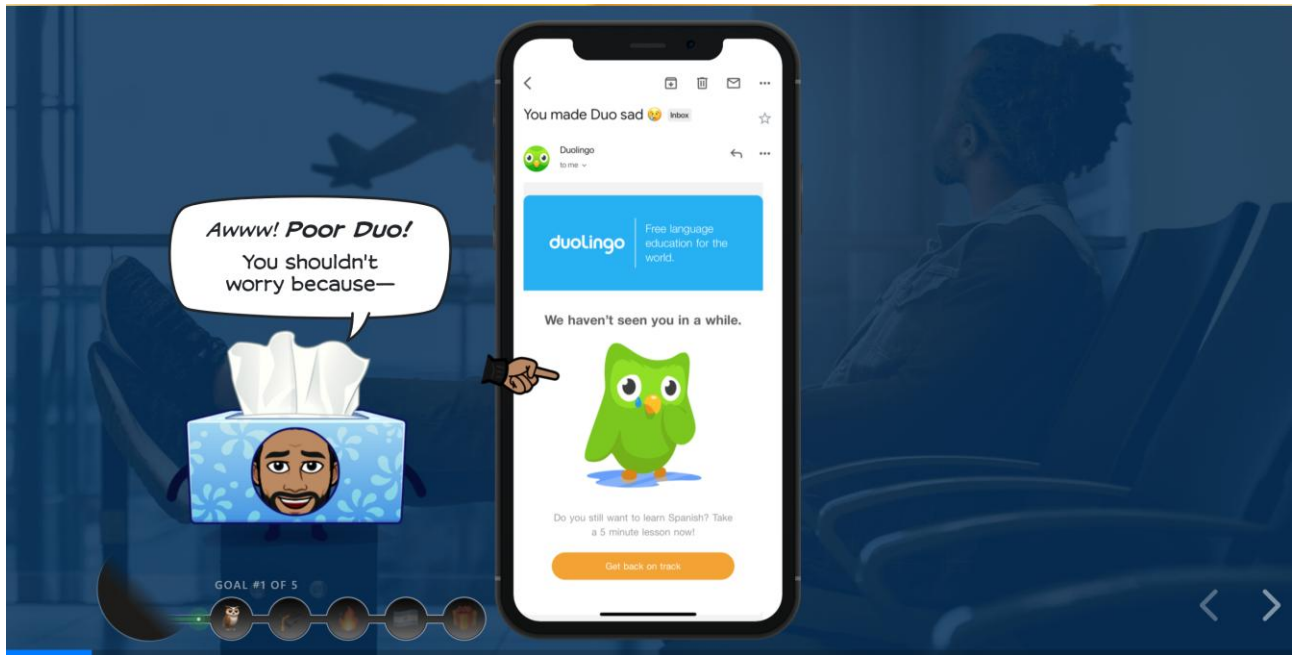


Figure 22 - Duolingo and the consistency principle (via [Growth Design](#))

Social proof principle

Whenever people are unsure about what to do, especially in a new environment, they stick to what most people around them are doing, following the social proof principle. Marketing, advertising, and user experience design use this principle every time something is described as “popular” or “the most chosen” option. However, as noted by Cardello (2014), social proof has many applications, such as reviews or social filters like “customer who viewed this also viewed”. The author suggests using this technique to increase credibility (if other people liked this content, it must be of quality) and to encourage adoption and acceptance, by highlighting, for instance, that a lot of people subscribed to your newsletter. Reviews are a typical application of this principle.



Figure 23 - [Unobravo](#) (french version of the website) and the social proof principle

Liking principle

According to the liking principle, we are more influenced by people that we like or that are similar to us. Despite this principle being more useful in marketing or advertising, there are some ways to implement this principle in user interface design. In another article, Cardello (2014) gives some examples of how to use the liking principle when designing interfaces, namely placing pictures of real people (such as the customer service staff or the company employees) and providing company history through the “about us” page. *Fiscozen* leveraged this principle in their “about us” page.



Figure 24 - *Fiscozen* and the liking principle

Authority principle

The authority principle explains that we are more likely to believe or be influenced by authorities. The widest application of this principle is the “approved by experts” used in advertising and marketing, along with the imagery of scientists with the white lab coat explaining why a certain product works. In user interface design authority can be represented by photos of people in authority positions (or dressed as if they were) or by symbols or authority, for example logos of reputable organizations (Fessenden, 2018).

An example can be found in *Pitch* homepage. Not only does it contain reviews from the CEO of *Perplexity*, but it also differentiates its reviews for different roles.

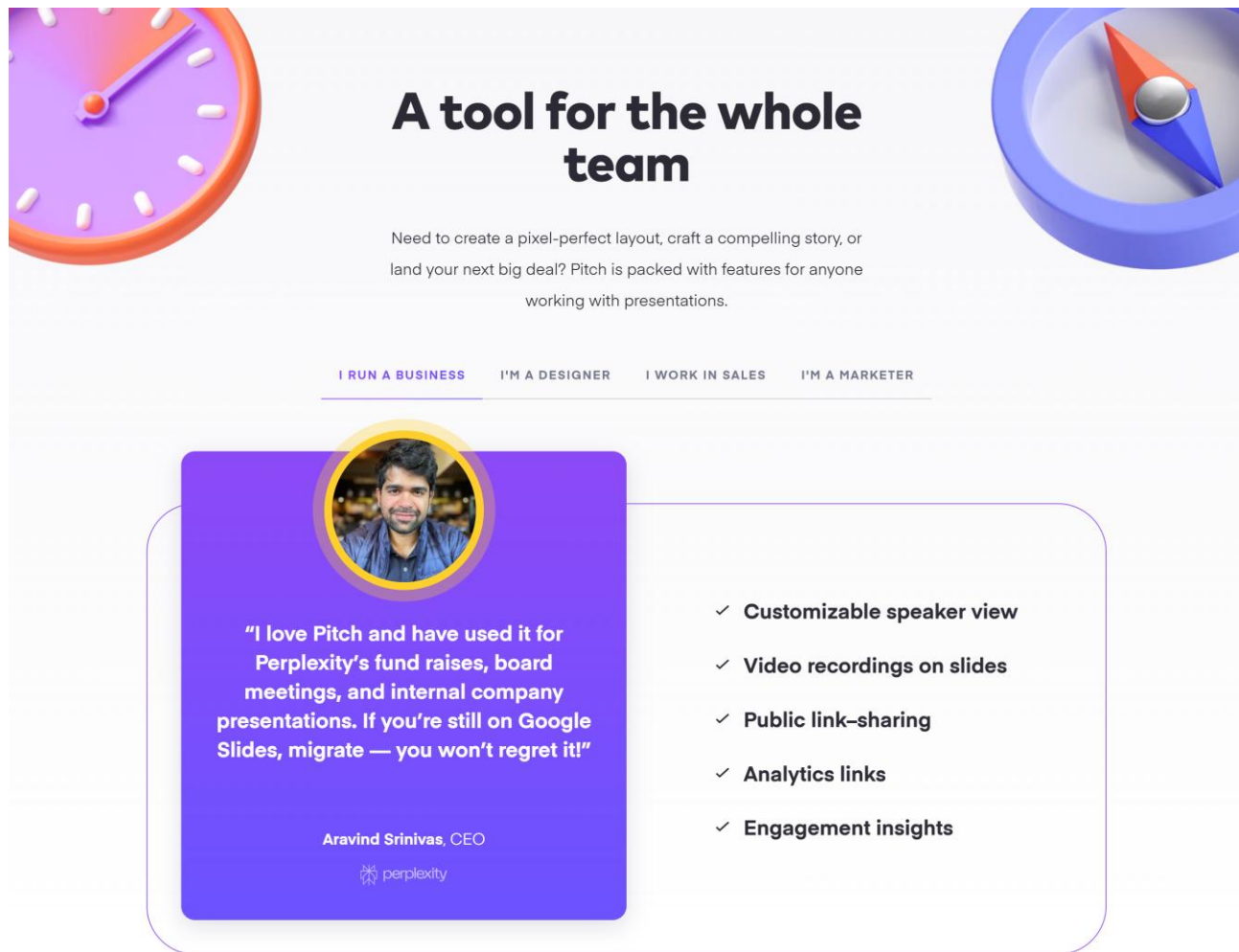
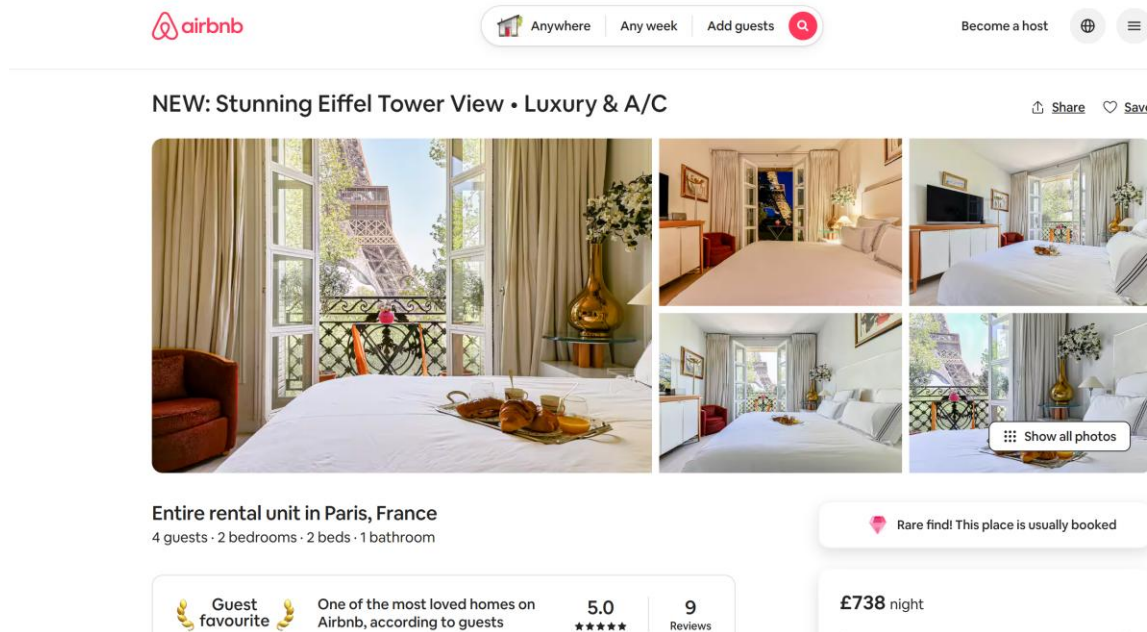


Figure 25 - *Pitch* homepage and the authority principle

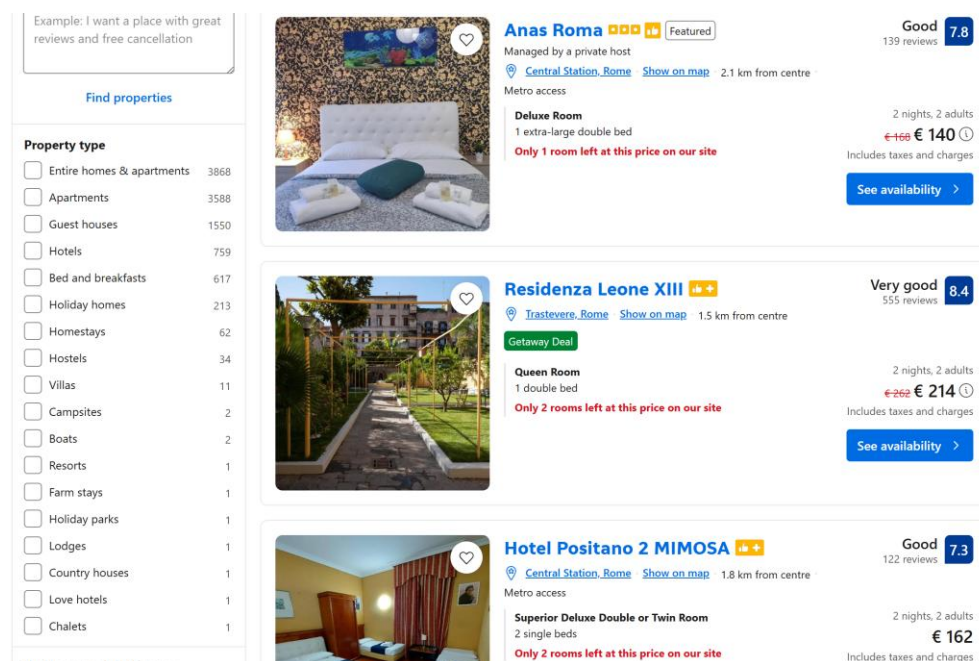
Scarcity principle

Finally, the most widely used among Cialdini's principles is scarcity. Whenever something is presented as limited, people perceive it as more valuable and tend to get it sooner than they would have had normally, for the psychological principles of reactance and loss aversion. As always, marketing, advertising, and user experience design exploit this principle. In an e-commerce or on every website where something can be bought, scarcity means presenting it as a limited-time offer, as something which is almost out of stock, or as something that other people are looking for in the present moment.

Figure 26 shows how *AirBnb* exploits this principle presenting this already stunning apartment as a "rare find", which is "usually booked", hence increasing the feeling of its rarity, and, consequently, its perceived value.

Figure 26 - [AirBnb apartment](#) and the scarcity principle

Another example comes from a similar company. *Booking* often conveys scarcity with the sentence “only 1 room left at this price on our site”, written in red to further signal its importance.

Figure 27 – [Booking](#) and the scarcity principle

Models of information elaboration in psychology of persuasion

One of the most successful models in psychology of persuasion is the Elaboration Likelihood Model or ELM (Cacioppo and Petty, 1984; Petty and Cacioppo, 1986). According to this model, recipients process a persuasive message along a continuum that goes from high

elaboration likelihood to low elaboration likelihood, following respectively a central or peripheral way of processing. Central processing leads to a more long-lasting shift in recipients' attitudes, while attitude change is more temporary when processing peripherally.

The likelihood of elaboration depends on cognitive resources availability, motivation to process the message (which depends on the personal involvement in the subject), time availability, and individual factors such as the need for cognition, which is the tendency to do and appreciate things that require mental effort. Whenever recipients have enough cognitive resources available and time to process the message, they are personally involved in the matter, and they are high in need for cognition, they are likely to process the message in a central way. Conversely, whenever people do not have enough cognitive resources and time, they are not personally involved, they are low in need for cognition, and they are not experts in the field, they are likely to process the message in a peripheral way.

Chaiken' (1980) heuristic/systematic model is similar to the ELM, with the heuristic and systematic way corresponding to the peripheral and central route of processing, respectively. Or, in the words of Kahneman (2011), System 1 and System 2.

When recipients are likely to elaborate the message centrally (or systematically), the source of the persuasive message must use high-quality arguments about the topic, whereas it should focus on peripheral cues like the number of arguments and the source attractiveness, similarity to recipients or expertise if recipients are likely to process the message peripherally (or heuristically).

As suggested by the Interaction Design Foundation (2015), designers should consider both the central/systematic way and the peripheral/heuristic way of elaboration in their products, because different people in different situations might have different levels of personal involvement, cognitive resources, time, and expertise. Hence, when designing a website or an application, it should be taken into account that high-quality arguments (like detailed information about the specifics of our product or service) must be accompanied by peripheral cues such as captivating media (images and videos), psychological principles about framing and pricing, or the attractiveness and expertise resulting from texts and visuals. *Dropbox* homepage tries to persuade prospects by using solid arguments (central route) and pleasing visual cues, such as color palette, animations, and media (peripheral route).

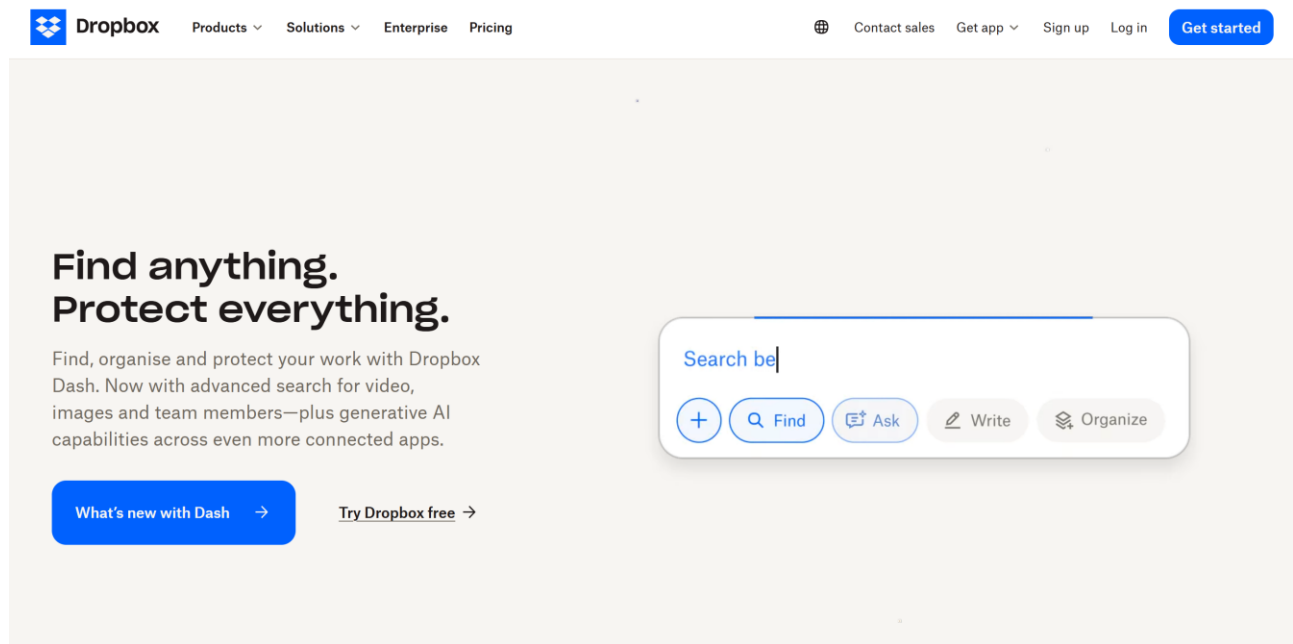


Figure 28 - [Dropbox](#) and the integration between central and peripheral route

CONCLUSIONS

Understanding users' needs and accommodating their way of thinking and behaving is essential for satisfying them and providing them with good user experience. Therefore, user interface designers must consider and exploit as many psychological principles as possible when conceiving and crafting their products. Most importantly, even though aesthetic-usability effect exists (Moran, 2024), designers must remember that user interfaces must be usable, before being aesthetically pleasing. These and other psychological principles should guide them in the design process.

AI

While writing this document, I have used [Perplexity](#) and [ChatGPT](#) to complement my bibliographic research. When I could not find out when a term or concept had been introduced, I have employed these tools, always double-checking the original source they referenced, as I know that sometimes they make things up.

Besides, since English is not my first language, I've used both tools, along with [Grammarly](#) and [Reverso Context](#), to check for grammatical mistakes or general improvement to the syntactic structure of what I have written down in the first place.

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